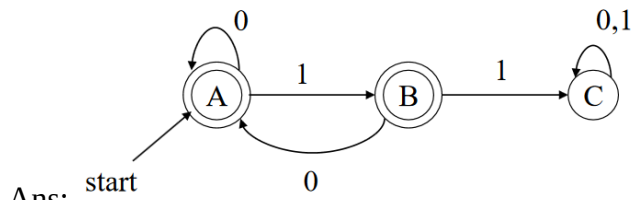


**UCS2603 THEORY OF COMPUTATION**  
**Answer key (CAT 1)**

**Part – A ( $4 \times 2 = 8$  Marks)**

1. Show a DFA which accepts all strings without two consecutive 1's over  $\Sigma = \{0,1\}$ .



Ans: start

2. Construct regular expressions for the following:  
Strings of a's and b's with no consecutive a's.

Ans

$(a+ε)(b+ba)^*$

3. **Select/ Find** is the type of grammar given by the following set of productions?

$A \rightarrow aA \mid bB$

$B \rightarrow bA \mid aB \mid \epsilon$

Ans: Regular Grammar (RG)

4. Find the language generated by

$S \rightarrow 0S1 \mid 0A \mid 0 \mid 1B \mid 1$

$A \rightarrow 0$

$B \rightarrow 1$

Ans: 0's followed by 1's

**Part – B ( $3 \times 6 = 18$  Marks)**

5. If L is accepted by an NFA then show that L is also accepted by a DFA.
6. Construct the regular expression whose language consists of strings with alternate 0's and 1's over an alphabet  $\Sigma = \{0,1\}$ .

Strings that will be acceptable by regular expression with alternate 0's and 1's –

1.  $\epsilon$  (no input, 0 and 1)

2. 010101..... (string that starts with 0 followed by 1 and so on).

3. 101010..... (string that starts with 1 followed by 0 and so on).

Now, a regular expression for set of all strings consisting of alternate 0's and 1's would be  **$(01)^*$** , where it can accept  $\epsilon$ , 01, 0101, 010101.....etc but this restricts the string as it can always begin with 0 only.

Again, the expression  $(10)^*$  will accept  $\epsilon, 10, 1010, 101010, \dots$  etc but this too restricts string as it can always begin with 1 only.

So, we introduce  $1(01)^*$  and  $0(10)^*$  to meet gap in respective cases.

While  $1(01)^*$  breaks the restriction of string starting with 0,  $0(10)^*$  breaks same for string starting with 1.

So, the final expression is –

$$(01)^* + (10)^* + 0(10)^* + 1(01)^*$$

7. State the pumping lemma for regular languages and check whether a language  $L = \{ww^r / w \in (a, b)^*\}$  is regular or not.

Ans:

Pumping lemma

Let  $L$  be a RL. Then there ~~exist~~ is a constant  $n$  such that for every string  $w$  in  $L$  such that  $|w| \geq n$ , we can break  $w$  into 3 strings  $w = xyz$ , where  $y \neq \epsilon$  i.e.  $|y| > 0$ ,  $|xy| \leq n$   $\forall i \geq 0$ ,  $xy^iz$  is also in  $L$ .

$w \in L$   
 $w = a^m b^m$   
 $w^r = b^m a^m$   
 $ww^r = a^m b^m b^m a^m$  any arbit. word.

$w = xyz$   
 $xy = a^p$   $|xy| \leq n$   
 $y = a^r$   $y \neq \epsilon$   
 $z = a^{m-p} b^m b^m a^m$   
 $xy^iz = xy y^{i-1} z = a^p a^{r(i-1)} a^{m-p} b^m b^m a^m$

$a^{m+r(i-1)} b^m b^m a^m$

$i=0$   $a^{m-r} b^m b^m a^m \notin L$

$i=1$   $a^m b^m b^m a^m \in L$

$i=2$   $a^{m+r} b^m b^m a^m \notin L$

$\therefore L$  is not regular.

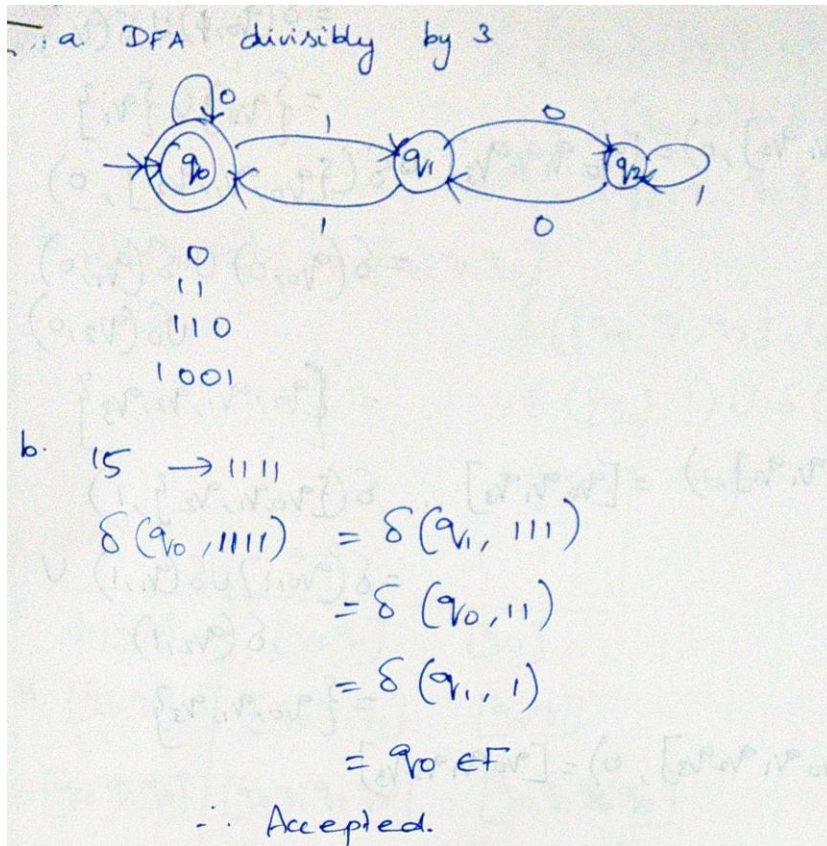
**Part – C (2 × 12 = 24 Marks)**

8. a. Construct a DFA for the following language L over the alphabet  $\Sigma = \{0, 1\}$  (8 Marks)

$L = \{w \mid w \text{ when interpreted as a binary number is divisible by 3}\}$

- b. Check the string validity for the number 15 using the constructed DFA.

Hint: Convert 15 to required format of the model (4 Marks)



9. a. Construct a DFA equivalent to the NFA.

$M = (\{p, q, r, s\}, \{0, 1\}, \delta, p, \{q, s\})$

Where  $\delta$  is defined in the following transition table.

| $\delta$ | 0     | 1     |
|----------|-------|-------|
| p        | {q,s} | {q}   |
| q        | {r}   | {q,r} |
| r        | {s}   | {p}   |
| s        | -     | {p}   |

**Solution :**

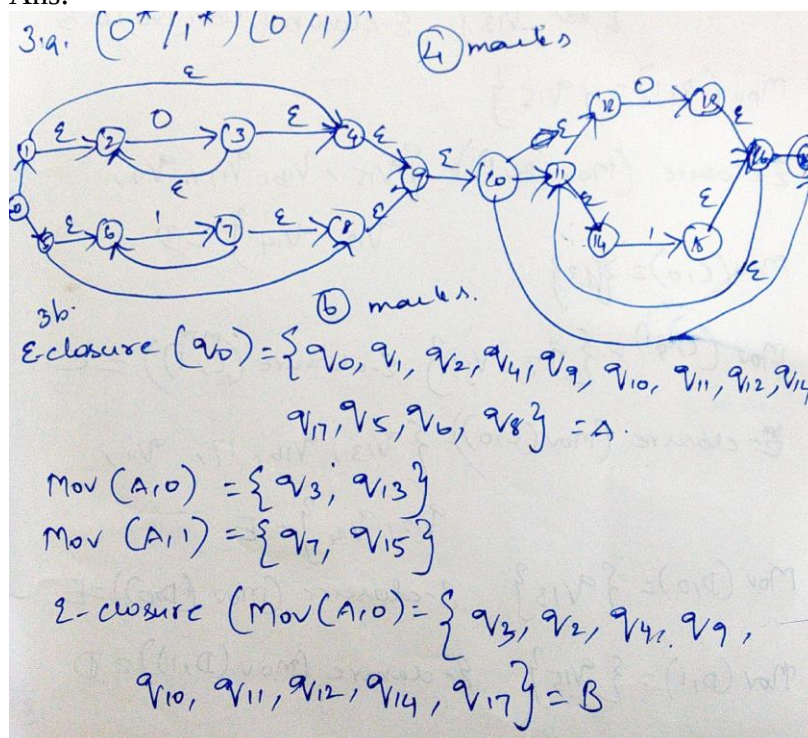
The equivalent DFA is as follows :

$$\begin{aligned}
 Q^1 &= 2^Q \\
 &= ([p], [q,s], [q], [r], [q,r], [s], [r,s], [p,q,r], [q,r,s]) \\
 \Sigma &= \{0, 1\} \\
 p &= \text{Initial state} \\
 F^1 &= \text{Final states} = ([q,s], [q,r,s], [q], [q,r], [s], [r,s], [q,q,r])
 \end{aligned}$$

| States    | Inputs    |           |
|-----------|-----------|-----------|
|           | 0         | 1         |
| $[p]$     | $[q,s]$   | $[q]$     |
| $[q]$     | $[r]$     | $[q,r]$   |
| $[r]$     | $[s]$     | $[p]$     |
| $[s]$     | $\phi$    | $[p]$     |
| $[q,s]$   | $[r]$     | $[p,q,r]$ |
| $[q,r]$   | $[r,s]$   | $[p,q,r]$ |
| $[r,s]$   | $[s]$     | $[p]$     |
| $[p,q,r]$ | $[q,r,s]$ | $[p,q,r]$ |
| $[q,r,s]$ | $[r,s]$   | $[p,q,r]$ |

10. Construct a minimized DFA for the RE  $(0^*+1^*)(0+1)^*$ .

Ans:



$$\epsilon\text{-closure}(\text{Mov}(A,1)) = \{q_7, q_{15}, q_8, q_9, q_{10}, q_6, q_{11}, q_{12}, q_{14}, q_{17}\} = C$$

$$\text{Mov}(B,0) = \{q_3, q_{13}\} \quad \epsilon\text{-closure}(\text{Mov}(B,0)) = B$$

$$\text{Mov}(B,1) = \{q_{15}\}$$

$$\epsilon\text{-closure}(\text{Mov}(B,1)) = \{q_{15}, q_{16}, q_{17}, q_{11}, q_{12}, q_{14}\} = D$$

$$\text{Mov}(C,0) = \{q_{13}\}$$

$$\text{Mov}(C,1) = \{q_7, q_{15}\} \quad \epsilon\text{-closure}(\text{Mov}(C,1)) = C$$

$$\epsilon\text{-closure}(\text{Mov}(C,0)) = \{q_{13}, q_{16}, q_{17}, q_{11}, q_{12}, q_{14}\} = E$$

$$\text{Mov}(D,0) = \{q_{13}\} \quad \epsilon\text{-closure}(\text{Mov}(D,0)) = E$$

$$\text{Mov}(D,1) = \{q_{15}\} \quad \epsilon\text{-closure}(\text{Mov}(D,1)) = D$$

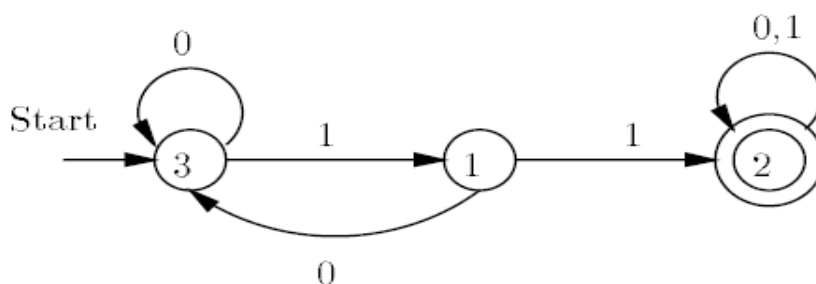
$$\text{Mov}(E,0) = \{q_{13}\} \quad \epsilon\text{-closure}(\text{Mov}(E,0)) = E$$

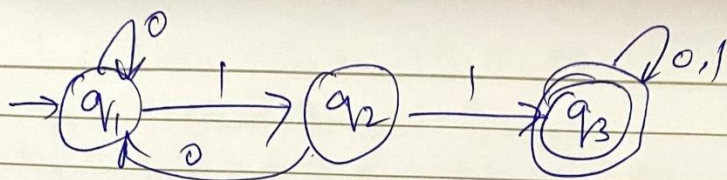
$$\text{Mov}(E,1) = \{q_{15}\} \quad \epsilon\text{-closure}(\text{Mov}(E,1)) = D$$

|    | 0 | 1 |
|----|---|---|
| A  | B | C |
| *B | B | D |
| *C | E | C |
| *D | E | D |
| *E | E | D |

Minimization 2 mark

11. Make use of Arden's Lemma to find the RE for the given DFA.





$$q_1 = q_1 \alpha_{11} + q_2 \alpha_{21} + q_3 \alpha_{31} + \epsilon$$

$$q_2 = q_1 \alpha_{12} + q_2 \alpha_{22} + q_3 \alpha_{32}$$

$$q_3 = q_1 \alpha_{13} + q_2 \alpha_{23} + q_3 \alpha_{33}$$

$$q_1 = q_1 0 + q_2 0 + q_3 \phi + \epsilon \quad \text{--- (1)}$$

$$q_2 = q_1 1 + q_2 \phi + q_3 \phi \quad \text{--- (2)}$$

$$q_3 = q_1 \phi + q_2 1 + q_3 (01) \quad \text{--- (3)}$$

Subs (2) in (3)

$$q_3 = q_1 1 + q_3 (01) \quad \text{--- (4)}$$

Sub (2) in (1)

$$q_1 = q_1 0 + q_1 10 + \epsilon$$

$$q_1 = q_1 (0+10) + \epsilon$$

$$R = 0+RP$$

$$R = 0P^*$$

$$q_1 = \epsilon (0+10)^*$$

$$\boxed{q_1 = (0+10)^*} \quad \text{--- (5)}$$

Substitute (5) in (4)

$$q_3 = \underbrace{(0+10)^*}_{Q} 1 + q_3 \underbrace{(01)}_{RP}$$

$$q_3 = (0+10)^* 1 (0+10)^* \leftarrow RE$$

$$q_1 = q_1 \alpha_{11} + q_2 \alpha_{21} + q_3 \alpha_{31} + \epsilon$$

$$q_2 = q_1 \alpha_{12} + q_2 \alpha_{22} + q_3 \alpha_{32}$$

$$q_3 = q_1 \alpha_{13} + q_2 \alpha_{23} + q_3 \alpha_{33} + \epsilon$$

$$q_1 = q_1 \phi + q_2 \phi + q_3 1$$

$$q_2 = q_1 1 + q_2 (0+1) + \phi$$

$$q_3 = q_1 0 + q_2 \phi + q_3 0 + \epsilon$$

$$q_1 = q_3 1 \quad \text{--- (1)}$$

$$q_2 = q_1 + q_2 (0+1) \quad \text{--- (2)}$$

$$q_3 = q_1 0 + q_3 0 + \epsilon \quad \text{--- (3)}$$

sub (1) in (3)

$$q_3 = q_3 (10+0) + \epsilon$$

$$R = R \quad P \quad + \quad Q$$

$$q_3 = \epsilon (10+0)^* \quad \text{--- (4)}$$

sub (4) in (1)

$$q_1 = (10+0)^* 1 \quad \text{--- (5)}$$

sub (5) in (2)

$$q_2 = (10+0)^* 11 + q_2 (0+1) \quad \text{--- (6)}$$

$$R = Q \quad R \quad P$$

$$q_2 = (10+0)^* 11 (0+1)^* \leftarrow RE$$